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Slow-light silicon modulator with 110-GHz bandwidth

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Silicon modulators are key components to support the dense integration of electro-optic functional elements for various applications. Despite numerous advances in promoting the modulation speed, a bandwidth ceiling emerges in practices and becomes an obstacle toward Tbps-level throughput on a single chip. Here, we demonstrate a compact pure silicon modulator that shatters present bandwidth ceiling to 110 gigahertz. The proposed modulator is built on a cascade corrugated waveguide architecture, which gives rise to a slow-light effect. By comprehensively balancing a series of merits, the modulators can benefit from the slow light for better efficiency and compact size while remaining sufficiently high bandwidth. Consequently, we realize a 110-gigahertz modulator with 124-micrometer length, enabling 112 gigabits per second on-off keying operation. Our work proves that silicon modulators with 110 gigahertz are feasible, thus shedding light on its potentials in ultrahigh bandwidth applications such as optical interconnection and photonic machine learning.

INTRODUCTION

Silicon modulators are the essential building blocks for silicon photonics (1) since they are responsible for converting electrical signals to optical ones, which are indispensable in any information technology applications such as data transmission, interconnection, processing, and computing (2–4). Leveraged by the complementary metal-oxide semiconductor (CMOS)-compatible feature of silicon photonics (5–7), silicon modulators are expected to support wafer-scale manufacturing, massive production, and low cost; thus, they are particularly promising in realizing next-generation optoelectronic applications in which an aggregate data throughput of Tbps-level is required on a single chip (8), and therefore, modulators with high bandwidth and compact size are dispensable in achieving a lane speed above 100 Gbps (9–12). However, despite extensive efforts, such as material engineering (13, 14), device optimization (15–18), and applying new photonic structures (19, 20), have paid to promote the operation speed of silicon modulators, the reported electro-optic (EO) bandwidth has been hindered at the level of about 60 GHz to date as a bandwidth ceiling (21–23), which raises concerns about the potential of silicon photonic toward ultrahigh-bandwidth applications. In particular, when looking at the advances that equal or higher bandwidth had been demonstrated upon heterogeneous materials (24–26) on silicon such as lithium niobate (27–31), polymer (32–35), and plasmonics materials (36–40), it raises a question that whether hybrid integration of a different material on silicon is a necessary approach for high-speed modulation, even in the cost of complex and CMOS-incompatible processes being involved. On the other hand, reducing the device dimension

is also critical for dense photonic integration (41, 42). Although using resonant or slow-light effects (43, 44) can markedly improve the modulation efficiency to enable compact size, it also brings a longer photon lifetime that may limit the bandwidth (45–47). Therefore, how to realize pure silicon modulators with ultrahigh bandwidth and compact footprint under CMOS-compatible manufacturing processes remains an important but elusive problem.

Here, we theoretically propose a strategy for realizing high-speed silicon modulators operating at telecom wavelength around 1550 nm by using the slow-light effect in an optical waveguide that is consisted of a series of cascaded Bragg gratings which is equivalent to coupled-resonator optical waveguide (CROW) in physics. By comprehensively balancing several factors including the group index, photon lifetime, electrical bandwidth, and losses, we design and fabricate the silicon modulators using the standard silicon photonic foundry workflow and demonstrate an ultrahigh bandwidth of 110 GHz in a length of 124 μm . To the best of our knowledge, this is a record-high bandwidth among the reported results, which breaks the bandwidth ceiling of silicon modulators to date. As confirmed by the transmission experiment, such a modulator is capable of supporting a data rate of 112 Gbps in a spectral window of 8 nm by applying simple on-off keying (OOK) modulation for eye diagram without applying any digital signal processing (DSP). Our findings pave the path for data communication beyond 110 Gbps per lane by using silicon photonics and highlight the potential of silicon photonics toward ultrahigh-bandwidth applications.

RESULTS

Design and principles

The goal of this work is to develop a pure silicon modulator by using the plasma dispersion effect (48) to achieve EO bandwidth in 100 GHz grade while keeping its footprint as small as possible. Therefore, the target modulator can support data transmission with a single lane speed above 100 Gbps by using simply OOK coding to best reduce the complexity and cost of DSP for short-range optic

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links and reserving the bandwidth budget for even higher lane speed by adopting high-order modulation format in the future. For this purpose, we found a carefully designed slow-light effect induced from a cascade corrugated waveguide that can effectively enhance the modulation efficiency while maintaining response speed without degradation.

In general, the EO bandwidths of silicon modulators (BW_{EO}) are determined together by their optical bandwidth (BW_O), which is characterized by the photon lifetimes (or equivalently, the quality factors Q s) of optical modes, and their electrical bandwidth (BW_E) that is governed by the RC constants of PN junctions and metal contacts, followed by (49)

$$\frac{1}{BW_{EO}^2} = \frac{1}{BW_E^2} + \frac{1}{BW_O^2} \quad (1)$$

To promote the overall bandwidth of modulators, it is essential to improve the optical and electrical bandwidths at the same time and find an appropriate balance between them. Noteworthy that, although it is well-known that strong slow-light effect could enhance the modulation efficiency, we found that a moderate value of group index is more favorable for achieving the balance between optical and electrical bandwidths.

We start with a schematic design of a fishbone-like waveguide as shown in Fig. 1A, which consists of a series of cascaded Bragg grating and acts as one arm of the silicon Mach-Zehnder modulator. Specifically, the light propagates along the waveguide in x direction, and the unit cell of the grating is composed of two fingers protruding in y direction with a lattice constant of a along x direction. Further, each number of period (N_p) of unit cells forms one arm of a resonator, and the two arms are separated by a $\lambda/4$ phase shifter region from their adjacent one. Therefore, the waveguide can be understood as a typical CROW with a finite number of cascaded resonators, counted by number of resonator (N_r).

First, we investigate the optical characteristics of the waveguide (using COMSOL Multiphysics and Lumerical). For the mentioned infinite grating, its bulk bands in the vicinity of X -point, namely, near $ka/2\pi = 0.5$, are presented in Fig. 1B, in which the antisymmetric and symmetric transverse electric (TE) bands are denoted as TE-A and TE-B with a bandgap Δ between them. The bandgap Δ depicts the coupling strengths between the counter-propagating waves in the grating and thus can be tuned by the geometry of the unit cell. When a finite number of unit cells are cascaded through the phase shifter region, it creates a supercell in the size of $2N_p$. Accordingly, TE-A and TE-B bands split into a set of discrete modes as reported (50–56), and we plot the lowest-order modes in Fig. 1C for TE-B (top) and TE-A (bottom) bands for examples. The phase shifter gives rise to a mid-gap mode embedded in the bandgap (Fig. 1B, red dot) with its electromagnetic field mostly localizing in the phase shifter region (Fig. 1C, mid panel). As discussed in the literature (57), such mid-gap modes in phase-shifted gratings are topological and are mathematically equivalent to the Jackiw-Rebbi zero mode (58).

Similar to the physics of CROW which has been investigated in report works (59, 60), both the discretized modes of bulk bands and mid-gap mode evanescently couple to the neighbor supercell in coupling coefficients $\kappa = \int d^3r [\epsilon_0(\mathbf{r}) - \epsilon(\mathbf{r})] \mathbf{E}_\Omega(\mathbf{r}) \cdot \mathbf{E}_\Omega(\mathbf{r} - 2N_p a \mathbf{e}_x)$, in which $\mathbf{E}_\Omega$ is the normalized electric field of modes. As a result, the couplings of neighbor supercell modes bring several

continuous supercell bands as we presented in Fig. 1D, which clearly exhibit the slow-light effect, measured by the group velocity $v_g = d\omega/dk$. Further, the supercell bands open several transmission windows in the spectrum as illustrated in Fig. 1E, and the spectrum width follows $\Delta\lambda \propto 2\kappa\lambda_0$, in which λ_0 is the central wavelength of the modes. We calculate the group index $n_g = v_g/c$ for the supercell band of mid-gap mode (Fig. 1F) and find that the group indices are keeping a relatively constant value of $n_g = 6.1$ near the center of the transmission window, which is suitable for the silicon modulator to operate thus we call it “passband.” Noteworthy to mention that, in such a slow-light waveguide, to avoid the transmission window being too narrow, the group index n_g cannot be too high, namely, the coupling coefficient κ cannot be too small.

To elaborate on the relationship and constraint between the design parameters, we simplify the coupling coefficient κ from evaluating the tunneling of the mid-gap mode through the bandgap, as

$$\kappa = \kappa_0 e^{-2N_p a \Delta} / N_p \quad (2)$$

where κ_0 is a constant determined by the unit cell structure; the gap depth Δ presented how strong the field is bound in the phase shifter region, and $2N_p a$ depicts the distance to tunneling through (see section S1 for the details). Then, the group index at the transmission window center can be explicitly written as $n_g = \lambda_0 / 4\pi\kappa N_p a = (\lambda_0 / 4\pi\kappa_0 a) e^{2N_p a \Delta}$ which is exponentially increasing with Δ and N_p , indicating a stronger slow-light effect.

It is well known that slow light is beneficial in enhancing the modulation efficiency of the silicon modulators that use the plasma dispersion effect (44, 61–64). The modulation in carrier concentration causes electrically controlled refractive index changes in silicon materials, resulting in an observable phase accumulation when light travels through. The details of modulators including the geometry parameters, doping level, and contact design are presented in Fig. 2A. In such a device, the phase accumulated in a length of $L = 2N_r N_p a$ is given by

$$\phi = \Delta n_{\text{eff}} \cdot L_{\text{eff}} \cdot \frac{2\pi}{\lambda_0} = \eta \cdot n_g \cdot (2N_r N_p a) \cdot \frac{2\pi}{\lambda_0} \quad (3)$$

where $\eta = \eta_0 (\sqrt{V + V_{bi}} - \sqrt{V_{bi}})$ with η_0 being a coefficient related to the overlapping between the carriers and optic field; V_{bi} is the built-in voltage of the PN junction, and V is the operating voltage of the modulator. Accordingly, we define the efficiency factor as $\phi/\pi V$ to characterize the phase change per voltage. The phase accumulation is enhanced by the slow-light effect in a factor of n_g . Given a determined n_g from Δ and N_p , the modulation efficiency linearly increases with N_r but shows square-root dependency on the voltage (see section S2 for the details).

We argue that the targeted optical bandwidth (BW_O) sets an upper bound of n_g and N_r . Specifically, the optical bandwidth can be depicted by the Q of mid-gap mode, given by $BW_O = 1/2\pi\tau = c/(\lambda_0 Q)$, in which τ is the photon lifetime and the Q are contributed by two parts as $Q^{-1} = Q_{\text{loading}}^{-1} + Q_{\text{propagating}}^{-1}$. Here Q_{loading} stands for the energy leakage at the ends of the slow-light waveguide, while $Q_{\text{propagating}}$ counts for the energy dissipation when the light propagates through the waveguide due to scattering loss and material absorption. Notice that $Q_{\text{propagating}}$ remains as a constant value with increasing waveguide length, and we estimate $Q_{\text{propagating}} \sim 5000$ from simulation. Moreover, Q_{loading} can be understood from the proportion between the total energy W stored in the waveguide

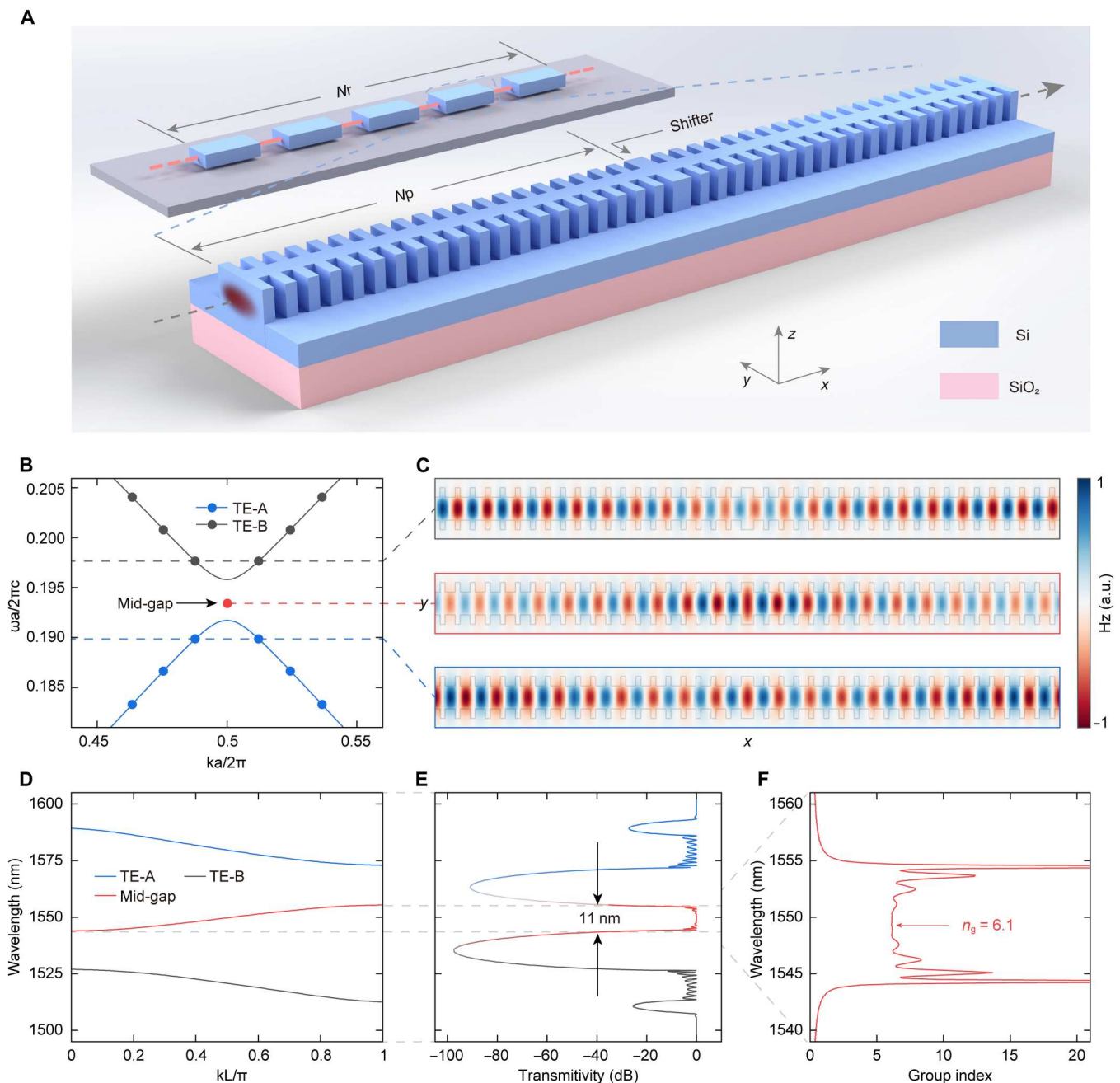


Fig. 1. Slow-light effect in cascade corrugated waveguide. (A) A schematic of the slow-light modulator on SOI wafer that consists of a series of fishbone-like Bragg gratings separated by phase shifter regions to form a slow-light waveguide. (B) The bulky band structure of the grating shows two bands TE-A and B in the vicinity of X point. When two neighboring gratings couple through the phase shifter, a topologically stable mid-gap mode emerges in the bandgap. (C) The magnetic field distributions (H_z) of lowest-order discretized bulk modes of TE-A (bottom) and TE-B (top) and the mid-gap mode (middle) in the slow-light waveguide supercell. a.u., arbitrary units. (D) The dispersions of three supercell bands correspond to modes TE-A, TE-B, and the mid-gap mode. (E) The transmission spectra of the supercell bands TE-A, TE-B, and the mid-gap mode, showing each of them open a transmission window. The spectrum width associated with the mid-gap supercell band is 11 nm. (F) The group index of the mid-gap supercell band indicates a modest slow-light effect of $n_g = 6.1$ at the center of the transmission window.

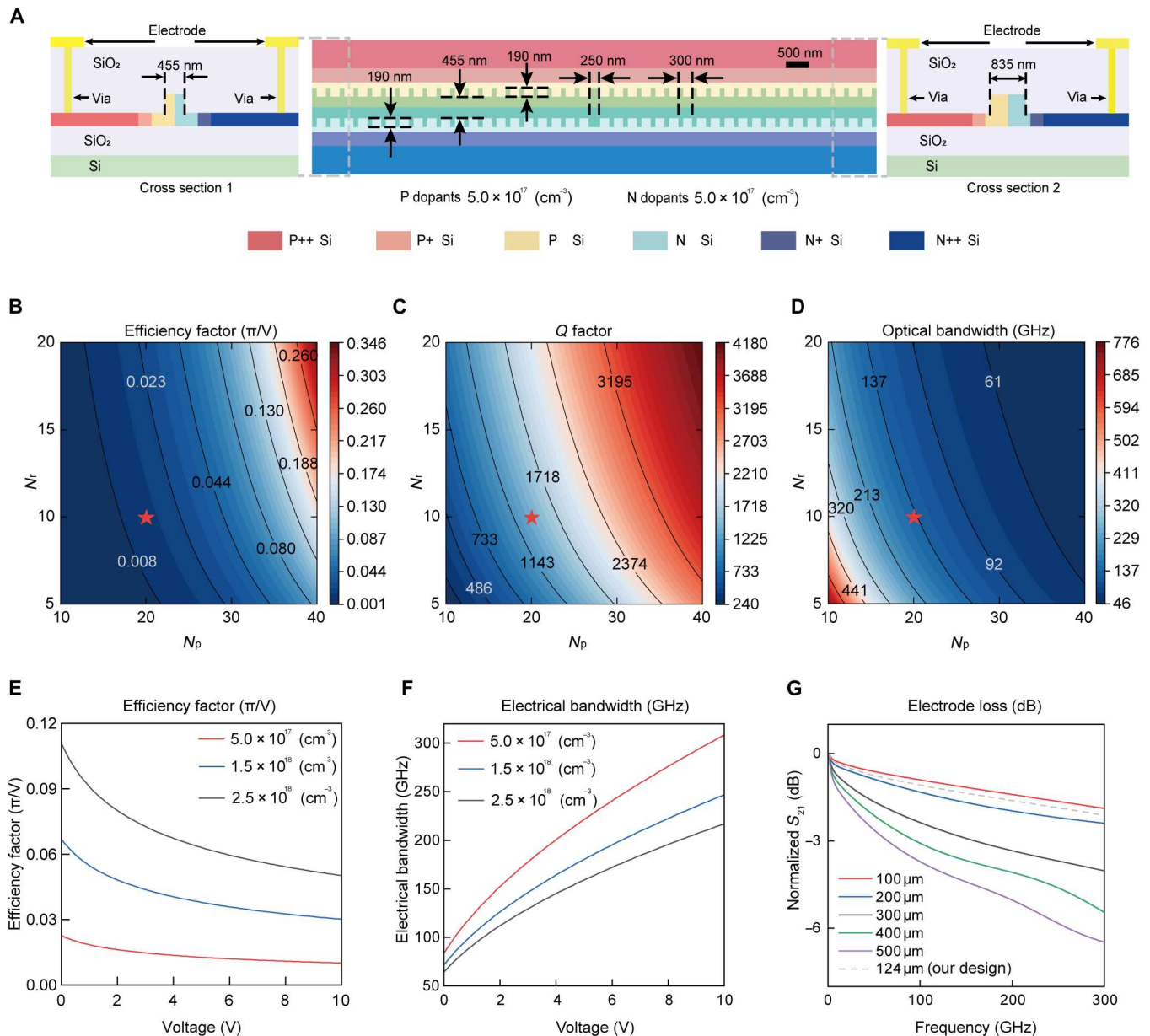


Fig. 2. Design of slow-light silicon modulator with ultrahigh bandwidth and compact size. (A) The detailed device geometry and doping configuration of the slow-light silicon modulator. (B to D) The maps of merits related to optical characteristics upon N_p and N_r , for (B) efficiency factor, (C) Q factors, and (D) optical bandwidth. (E and F) The electrical merits of the silicon modulator under different doping levels, for (E) efficiency factor and (F) electrical bandwidth. (G) Normalized electrode loss at different modulation frequencies and device lengths, indicating a marked loss increment when the device length exceeds 200 μm . The star marks show our optimal design.

and the energy flux P_{loading} escaping from its ends. We calculate the maps of efficiency factor, Qs, and optical bandwidth on varying N_p and N_r (see section S3 for the details), as illustrated in Fig. 2 (B to D). Noteworthy that there is a trade-off between modulation efficiency and optical bandwidth. To guarantee the high optical bandwidth to support a silicon modulator with 110 GHz, we found that $BW_O \sim 140 \text{ GHz}$ is an appropriate value, corresponding to $n_g = 6.1$ and efficiency factor of $0.013 \pi/V$ (star marks). Despite such an efficiency factor representing a relatively large V_π value (mostly due to the nonlinear dependency between the voltage and phase; see section

S2 for the details), sufficient modulation depth can still be achieved by taking the advantage of the linear modulation region with a proper reverse bias voltage.

We further discuss the electrical bandwidth (BW_E) and modulation efficiency. First, the doping level of the PN junction would influence such two attributes in a contrary way. As presented in Fig. 2 (E and F), a higher doping concentration promotes the efficiency factor under the same voltage, but considering that the junction capacitance also significantly increases with heavier doping for a modulator under depletion scheme, the electrical bandwidth, which is

calculated by $BW_E = 1/2\pi R_f C_p$, decreases accordingly. Therefore, we choose a relatively lower doping level of $5.0 \times 10^{17}/\text{cm}^3$ for our design, which gives $BW_E \sim 200$ GHz at 4 V voltage, to support the overall BW_{EO} of 110 GHz.

An operational modulator must accumulate a large enough phase in a limited length to achieve sufficient intensity modulation depth through Mach-Zehnder interference. As shown in Eq. 3, it seems that simply lengthening the slow-light waveguide with large N_r can always realize any desired modulation efficiency. However, other factors including the phase mismatching between optical and microwave velocities, optical loss of the slow-light waveguide, and electrical loss of the electrodes would set an upper limit of the total length. As discussed in section S4, we found that the loss of contacts is the bottleneck among many factors (65). We calculated the normalized S_{21} responses for a series of modulator lengths by applying the data estimated from our previous experiments (Fig. 2G). We found that when the modulator length is longer than 200 μm , the electrical loss markedly increases to an unacceptable level when operating at 200 GHz. Therefore, the limit length of the modulator results in a lower bound of n_g , namely, the slow-light effect cannot be too weak. Through validating the mentioned constraints, we confirmed that a modest $n_g = 6.1$ is sufficient to accumulate an operational phase in a length of ~ 150 μm .

To summarize the design strategy, the key to realize high-speed silicon modulators in a limit length is to find an appropriate group index n_g by designing the unit cell geometry of slow-light waveguide, which opens a wide enough transmission window of mid-gap mode but does not limit the optical bandwidth to coordinate with electrical bandwidth; on the other hand, sufficient large n_g is required to accumulate phases in a limit length. Therefore, a silicon modulator with 110 GHz is feasible by comprehensively balancing the mentioned factors.

Sample fabrication

The proposed strategy offers a guideline to find the optimal design parameters of a silicon EO modulator working under the cascade corrugated waveguide scheme, with a compromise in high bandwidth and efficiency. To verify the theory, a silicon modulator targeting a bandwidth of >100 GHz is fabricated from commercial silicon photonic foundry. The grating period is designed to be 300 nm, with a side wall corrugation depth of 190 nm. The N_p and N_r are selected as 20 and 10, respectively [starred point in Fig. 2 (B to D)]. All the above feature sizes are feasible for the standard foundry-based process (more fabrication details could be found in Materials and Methods). The visual morphology of the fabricated device under an optical micrograph is shown in Fig. 3A. Compared with the conventional silicon modulator, the footprint is much more compact which is over an order of magnitude shorter (in a length of ~ 124 μm). Figure 3B illustrates the scanning electron microscopy (SEM) picture of the zoom-in top view of the fabricated waveguide, showing that the grating teeth become slightly rounded due to fabrication imperfection which is tolerable for the designed performance.

Experimental characterization

To characterize the performance of the fabricated silicon modulator, we construct a measurement system as schematically shown in Fig. 4A. A tunable continuous wave laser incidence and radio frequency (RF) signal are applied to the silicon modulator to drive the

modulator for measuring the optical and electrical bandwidth. For evaluating the performance of high-speed data transmission, an oscilloscope is used to monitor and collect the data, to quantitatively obtain the optical modulation amplitude (OMA), signal-to-noise ratio (SNR), and extinction ratio (ER) of the optical link. See sections S7 to S9 for the details of specific measurements.

The optical loss of the modulator was extracted from the power difference between the coupling fiber, excluding an unoptimized coupling loss of ~ 10 dB, which is caused by a pair of grating couplers with 5-dB coupling loss for each (corresponding to a coupling efficiency of 31.6%) that works as input and output end with the optical fibers. The insertion loss of the 124- μm modulator is then measured to be 6.8 dB, incorporating 5.4-dB loss from the phase shifter and remaining from other components (e.g., directional couplers, routing waveguides, etc). Although the loss per unit length is a bit larger, the overall modulator loss is still at the same level of a standard 3-mm silicon modulator operated under depletion mode that fulfills the requirement for practical deployment.

Figure 4B shows the measured optical spectrum of the modulator around 1.55 μm , in which several transmission windows of mid-gap mode and other discretized modes of bulk bands are observed. Large attenuation on both sides of the spectrum is mainly due to the bandwidth limitation of the grating couplers we adopted as input/output optical ports. The passband (shading area of mid-gap mode) around 1550 nm holds an out-of-band rejection ratio of ~ 55 dB in a flat optical window width of ~ 8 nm (Fig. 4C), showing great consistency with the design (red lines). Such a device exhibits a much larger EO bandwidth than ever before among those reported pure silicon modulators. As shown in Fig. 4D, the frequency response is flat over the entire 110 GHz measurement range, without fast roll-off of bandwidth degrading. The 3-dB EO bandwidth is read to be about 110 GHz from fitting the frequency response curve, which is slightly degraded due to the intrinsic loss of the RF probe. Moreover, the remarkably high electrical bandwidth helps to eliminate the signal impairment when applying a high-speed driving signal, especially for the serial modulation format such as OOK, in which the Nyquist bandwidth is doubled compared with its 4-level pulse amplitude modulation (PAM-4) counterpart. Therefore, a 112-Gbps OOK eye diagram could be real time recovered without any pre-equalization at the transmitter side (see Fig. 4E), bringing the benefits in a reduction of the latency budget, as well as lowering the power consumption of the DSP.

The compact footprint and the large EO bandwidth, together with the wide optical operation window, make the modulator suitable for dense wavelength division multiplexing (DWDM)-based optical communication applications. A multiwavelength data transmission experiment using the fabricated modulator is carried out and presented in Fig. 4F. As proof, we perform the high-speed optical signal generation at a series of wavelengths within the 8-nm passband, as shown in Fig. 4G. The results show great consistency in each wavelength, confirmed by the measured variations in OMA, SNR, and ER, which are smaller than 0.5 dB across the passband.

We then perform the bit error rate (BER) measurement under different data rates. As shown in Fig. 4H, we measured the BER at different working wavelengths, with the OOK signal speed set as 70, 84, 98, and 112 Gbps. A slight performance fluctuation could be found, which may be due to the nonflatten gain spectrum of the pre-amplifier at receiving end. Noteworthy is that here the SNR

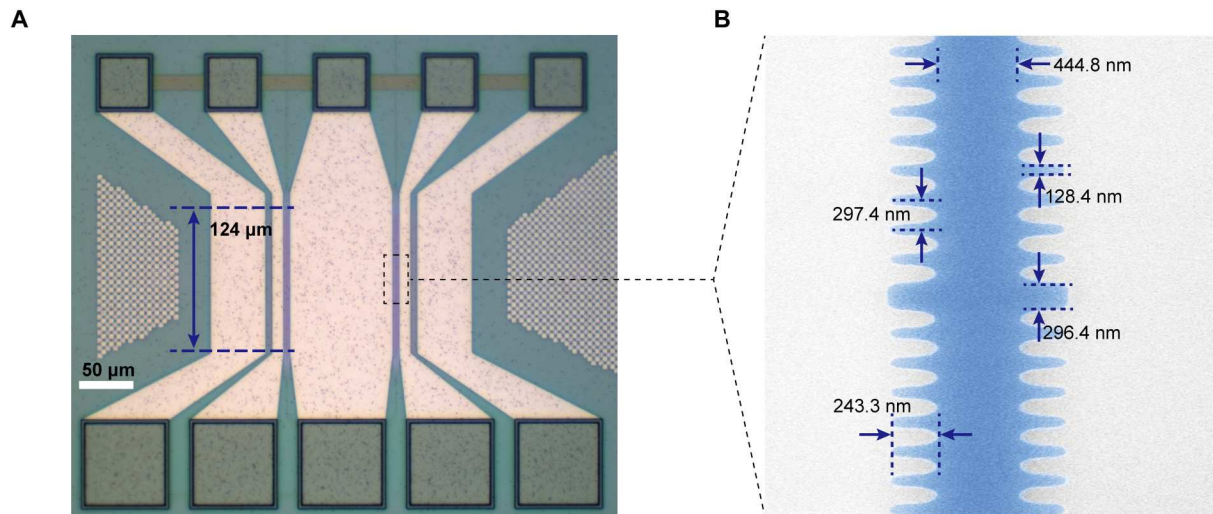


Fig. 3. Fabrication of the designed slow-light silicon modulator. (A) Optical micrograph of the fabricated slow-light silicon modulator. The modulation region has an ultracompact footprint, consisting of two 124-μm-long, PN-doped slow-light waveguide arms to form the MZI configuration. (B) The zoom-in SEM image shows part of the corrugated waveguide, which is composed of a $\lambda/4$ phase shifter and tens periods of fishbone-like gratings on both sides. Some critical dimensions of the fabricated device (e.g., the core width of the waveguide, the grating spacing, the width of the phase shifter region, etc.) are measured and marked.

and BER is not strictly corresponding to each other mainly because the PDs with different spectral responses are used in the eye diagram and BER tests, along with an extra erbium-doped fiber amplifier (EDFA) is adopted in the eye diagram setup. Figure 4I further illustrates the BER curves under the same data rates with the variations of the receiving power. With a proper SNR, the BERs can drop well below the hard-decision forward error coding (HD-FEC) threshold (3.8×10^{-3}) under the data rate of 98 Gbps and below the soft-decision forward error coding (SD-FEC) threshold (2×10^{-3}) when up to 112 Gbps (~ 93 -Gbps net rate).

DISCUSSION

The demonstration of a slow-light silicon modulator with an unprecedentedly high EO bandwidth of 110 GHz and an ultracompact length of 124 μm may broaden the horizon of silicon photonics. Both the bandwidth and the footprint of the device, to the best of our knowledge, break the performance limitation of a pure silicon modulator in traditional cognition, without additional complicated processes or aids from heterogeneous functional materials. Also, the work performed the first DSP-free data transmission experiments with a symbol rate of 112 GBaud identified by the eye diagram test, showing its potential in the next-generation datacom and telecom applications with the requirements of low latency and power consumption. Compared with the other type of resonant modulator, microring/disk modulator (41, 45, 66), an optimal designed cascaded corrugated waveguide modulator exhibits a compact size in the same scale which is more comparable in size to their electronic supporting elements meanwhile holding wider optical bandwidth and being less sensitive to temperature variation, which helps make full use of the spectral resources and save the power budget, benefiting the applications such as optical neural network and highly parallel data transmission, for high-volume throughput and large-scale integration (2). Despite a full π shift requiring a relatively large driving range due to the nonlinear

relationship between the applied voltage and phase changing in the slow-light waveguide, the balanced design ensures that the device can use the linear phase changing region to support sufficient modulation depth for the lane speed larger than 100 GBaud.

Following the proposed guideline, the silicon modulator could be flexibly designed to target different specifications and scenarios. In our OOK demonstrations of the eye diagram, where the device bandwidth is more critical to realize a DSP-free high-speed data transmission, comprehensive optimization of design parameters has been conducted from the proposed guidelines to ensure an ultralarge bandwidth, while the designed driving capability is sufficient of supporting two-level OOK signal transmission. Other advanced modulation formats, such as PAM-4 and 16-ary quadrature amplitude modulation (16-QAM), could further be used with an increased modulation depth by remodifying the design parameters (e.g., with a larger resonator number N_r) for better separation of the multilevel signals, in which cases the aggregation rate of a single carrier could potentially raise up above 1 Tbps. Besides the slow-light waveguide structure, the high-frequency electrode is another degree of freedom for high-speed design, which has not been specifically optimized in this work. Therefore, if the RF loss is further reduced by the substrate removed process (18) or the relative match condition is delicately managed (67), then the frequency response of the proposed structure is expected to be even higher. Moreover, the proposed design strategy can also be applied to other integrated platforms, such as III-V or LNOI (30), where the footprint is a long-term roadblock for high-density integration and a sophisticatedly regulated slow-light effect would help.

To summarize, we propose and implement a design strategy for silicon modulators by using a comprehensively balanced slow-light effect in cascade corrugated waveguide to achieve an EO bandwidth with 110 GHz in a compact length of 124 μm. The group index of cascaded mid-gap mode is optimized to a modest value of $n_g = 6.1$ by trading-off the passband width, optical bandwidth, modulation efficiency, and best coordinates with electrical bandwidth. By using

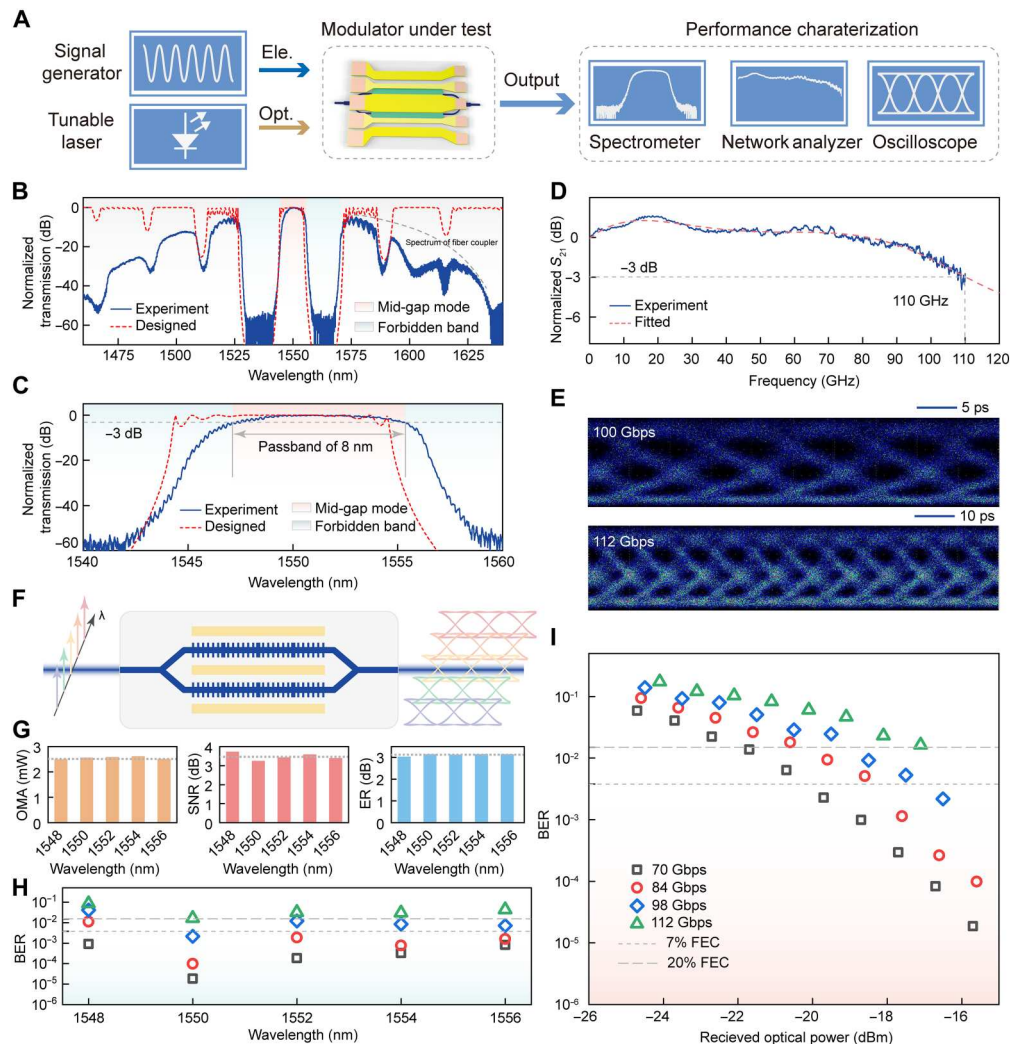


Fig. 4. Characterization of the ultrahigh bandwidth and ultrahigh-speed data transmission. (A) The schematic of the measurement system to characterize the fabricated slow-light silicon modulator. (B) The measured transmission spectrum (blue curves) in a range from 1460 to 1640 nm, which shows great consistency with the design (red lines). (C) The transmission spectrum of the mid-gap supercell band, namely, the passband, resides in the vicinity of 1550 nm. The passband shows as a flat-top window in a spectrum width of 8 nm with 55 dB out-of-band rejection ratio. (D) The measured (blue) and fitted (red) S_{21} curves of the EO response of the slow-light silicon modulator confirm its 3-dB bandwidth reaches 110 GHz. (E) The DSP-free optical eye diagrams for the OOK modulation at the data rates of 100 Gbps (top) and 112 Gbps (bottom), respectively. The measured ER is 3.15 and 2.15 dB, respectively. (F) The scheme of the multiwavelength data transmission using the fabricated slow-light silicon modulator. (G) The OMA, SNR, and ER performance across the operational passband of 8 nm. (H) The consistency of the measured BERs across multiple wavelengths in the passband under OOK modulation, in data rates of 70, 84, 98, and 112 Gbps. (I) The measured BER versus received optical power under a variety of data rates from 70 to 112 Gbps. FEC, forward error coding.

the fabricated modulator, we experimentally demonstrate a data rate of 112 Gbps in an operational window width of 8 nm by applying simple OOK modulation, which validates the proposed design strategy. Our work breaks the present bandwidth ceiling of silicon modulators and reveals the great potential of silicon photonics for the next-generation high-speed data transmission, wide-band signal processing, and large-volume photonic computing.

MATERIALS AND METHODS

Design and fabrication of the devices

The silicon modulator was fabricated on a 200-mm silicon on insulator (SOI) wafer with a silicon-layer thickness of 220 nm and a

buried oxide layer thickness of 2 μm using a standard 90-nm lithography SOI process at CompoundTek Pte in a one-to-one 200-mm-wafer run. The width of the waveguide core in the fishbone grating structure is 455 nm, with a 90-nm-thick partially etched rib area for carrier doping and metal contact. The concentrations of P-type and N-type doping at the waveguide core are $5.0 \times 10^{17}/\text{cm}^3$ and $5.0 \times 10^{17}/\text{cm}^3$, respectively. Meanwhile, the doping concentrations of intermediate P+ and N+ regions are $2.0 \times 10^{18}/\text{cm}^3$, and the P++ and N++ regions have the concentrations of $4.0 \times 10^{20}/\text{cm}^3$ for ohmic contact. To ensure that the waveguide mode adiabatically transit into the slow-light mode, waveguide tapers are used to connect the single mode strip waveguide (450 nm) and the phase shifting region. A pair of C-band grating couplers is used for light input/

output. As for the electrical part, the modulator is designed to work under the push-pull driving configuration. The push-pull operation mode can further reduce the chirp generation produced by the absorption effect of silicon simultaneously. Thus, a GSGSG-type high-frequency electrode is used here. To realize the characteristic impedance match-up, the gap between the ground and signal electrode is 6.4 μm , with the Cu electrode thickness of 1.2 μm .

Experimental details

For the bandwidth test, a vector network analyzer (Keysight PNA-X Network Analyzer N5247B) with its plug-in 110-GHz lightwave component analyzer (Keysight LCA Optical Receiver N4372E) is used to measure the linear EO transmission. The connection RF wires hold an electrical bandwidth of 100 GHz and are precalibrated over 110 GHz before the test. In the data transmission experiments, the high-speed OOK signal with a standard pseudo-random binary sequence (PRBS) pattern is generated by a bit pattern generator (SHF 12104A), and then the signal speed is doubled to over 100 Gbaud by a MUX (SHF C603B). The signals are amplified by a commercial driver (SHF S807C) to obtain a 5-V V_{pp} and then injected into the modulator working under the push-pull configuration. At the receiving end, eye diagrams are produced by a sampling oscilloscope (Tektronix DSA8300). As for the bit error ratio measurement, the same configuration but an arbitrary wave generator (Keysight M8199A) is adopted for programmable signal generation. After data receiving, the signals are fast sampled by a real-time oscilloscope (Keysight UXR0594AP) and afterward offline processed with feed-forward equalization (FFE) and maximum-likelihood sequence estimation (MLSE) algorithms. The receiving power is adjusted by a variable optical attenuator before the pre-amplifier (Amonics AEDFA-PA-35-B-FA).

Supplementary Materials

This PDF file includes:

Sections S1 to S9

Tables S1 and S2

Figs. S1 to S17

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